

Delay-Aware Rendezvous of Unicycle Multi-Robot Systems

Communication delays, nonholonomic constraints and the role of the critical delay τ_{crit}

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Control of Multi-Robot Systems — Project 2026

Abstract

This report studies the *rendezvous* problem for a team of nonholonomic unicycle robots that exchange position information over a communication graph affected by a transport delay τ . The work combines two complementary viewpoints. First, we use the classical linear full-state delayed-consensus model to derive the critical delay

$$\tau_{\text{crit}} = \frac{\pi}{2k\lambda_{\max}(L)},$$

which is an exact stability threshold for the corresponding single-integrator consensus system. Second, we use this value as a theoretical reference for the more realistic nonholonomic unicycle system, where the desired consensus velocity cannot be applied directly because each robot is constrained to move along its heading direction.

For this reason, τ_{crit} should not be interpreted as a strict necessary and sufficient stability boundary for the complete nonlinear unicycle dynamics. Rather, it provides a meaningful baseline: it captures the delay margin of the underlying communication-consensus layer and allows the simulations to be normalised through the ratio τ/τ_{crit} . The numerical experiments show that the unicycle rendezvous behaviour changes significantly around this theoretical threshold, while the precise transition is affected by nonholonomic constraints, heading dynamics, velocity saturation and the specific controller used to project the consensus field onto admissible unicycle motions.

The study validates this interpretation through delay sweeps, topology and gain studies, disconnected graphs, and several extensions such as leader–follower rendezvous, rigid formations, obstacle avoidance, switching graphs, time-varying delays and edge-dependent delays. The full-state critical-delay result is the gain-scaled form of the classical uniform-delay consensus bound established by Olfati-Saber and Murray [1]. The neighbour-only communication-delay model is also related to the framework studied by Seuret, Dimarogonas and Johansson [2], while the nonholonomic rendezvous controller is motivated by the consensus-based rendezvous and formation laws for unicycle robots proposed by Listmann, Masalawala and Adamy [3].

1 Introduction

Rendezvous is one of the canonical problems in multi-robot coordination: a group of agents, each measuring only relative information with respect to a set of neighbours, must agree on a common meeting point using a distributed control law. When the agents are simple integrators and information is exchanged instantaneously, the classical linear consensus protocol drives the team to the centroid of the initial positions. Two features make the present study more realistic and more challenging.

First, **communication is delayed**. Sensing, packetisation, transmission and processing all introduce a latency τ between the instant a neighbour’s position is generated and the instant it is used in the control law. Delay is not a benign perturbation: beyond a precise threshold it destabilises the very protocol that would otherwise guarantee convergence. A central goal of this work is to derive that threshold, the *critical delay* τ_{crit} , and to explain its dependence on the controller gain and on the spectrum of the graph Laplacian.

Second, the agents are **unicycles**, not integrators. The nonholonomic constraint forbids instantaneous motion in an arbitrary direction: each robot can only move along its current heading and steer. The consensus “force” produced by the delayed protocol must therefore be reconciled with the admissible motion of the vehicle, which introduces a nonlinearity absent from the linear reference model. This is

why the nonholonomic part of the project is treated as a controlled realisation of the consensus field, in line with consensus-based rendezvous controllers for unicycle robots [3].

The report is organised as follows. Section 2 fixes the unicycle model and the graph-theoretic notation. Section 3 derives τ_{crit} from a frequency analysis of the delayed consensus protocol and discusses its meaning. Section 4 compares the two delay models (full-state and neighbour-only). Section 5 explains how the consensus field is realised on the unicycle. Section 6 presents the numerical validation, and Section 7 the extensions. Section 8 concludes.

2 Model and preliminaries

2.1 Unicycle kinematics

We consider N unicycle robots. The state of robot i is the pose (x_i, y_i, θ_i) , with planar position $p_i = (x_i, y_i) \in \mathbb{R}^2$ and heading θ_i . The kinematics are

$$\dot{x}_i = v_i \cos \theta_i, \quad \dot{y}_i = v_i \sin \theta_i, \quad \dot{\theta}_i = \omega_i, \quad (1)$$

where v_i is the forward (linear) velocity and ω_i the angular velocity. The pair (v_i, ω_i) is the control input, subject in the implementation to the saturation $|v_i| \leq v_{\max}$, $|\omega_i| \leq \omega_{\max}$.

The nonholonomic constraint is made explicit by writing the velocity of the contact point as $\dot{p}_i = v_i t_i$, with $t_i = (\cos \theta_i, \sin \theta_i)$ the unit tangent. The robot cannot move along the normal direction $n_i = (-\sin \theta_i, \cos \theta_i)$ instantaneously; lateral displacement is only achievable by first steering. This is the key structural difference with respect to a single integrator $\dot{p}_i = u_i$, which can move in any direction at will.

2.2 Communication graph

Information exchange is described by an undirected weighted graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, A)$, with $\mathcal{V} = \{1, \dots, N\}$, edge set $\mathcal{E}$ and symmetric adjacency matrix $A = [a_{ij}]$, $a_{ij} = a_{ji} \geq 0$, $a_{ii} = 0$. Robot j is a neighbour of i (written $j \in \mathcal{N}_i$) iff $a_{ij} > 0$. The degree matrix is $D = \text{diag}(d_1, \dots, d_N)$ with $d_i = \sum_j a_{ij}$, and the graph Laplacian is

$$L = D - A. \quad (2)$$

For an undirected graph L is symmetric positive semidefinite, so its eigenvalues are real and can be ordered as

$$0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N = \lambda_{\max}(L).$$

The all-ones vector $\mathbf{1}$ is always an eigenvector associated with $\lambda_1 = 0$, because each row of L sums to zero. The second smallest eigenvalue λ_2 is the *algebraic connectivity*: $\lambda_2 > 0$ if and only if the graph is connected, and its magnitude quantifies “how connected” the graph is, governing the asymptotic convergence rate of consensus. The largest eigenvalue $\lambda_{\max}$, as we will see, governs the delay margin.

Throughout the baseline experiments we use a ring graph on $N = 6$ nodes (Figure 1), for which $\lambda_2 = 1$, $\lambda_{\max} = 4$.

3 The delayed consensus protocol and the derivation of τ_{crit}

3.1 Linear consensus without delay

We begin with the linear reference model in which each agent is a single integrator, $\dot{p}_i = u_i$, and the control is the standard consensus law

$$u_i = -k \sum_{j \in \mathcal{N}_i} a_{ij} (p_i - p_j), \quad k > 0. \quad (3)$$

Stacking the positions into $p = (p_1^\top, \dots, p_N^\top)^\top$ and using (2), the closed loop reads, componentwise in each Cartesian coordinate,

$$\dot{p} = -kLp. \quad (4)$$

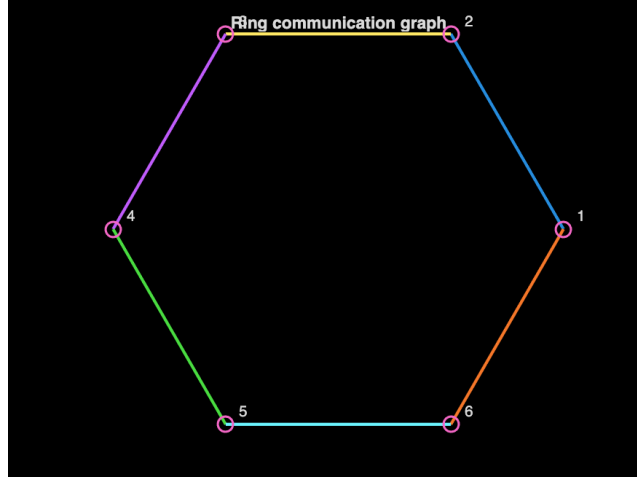


Figure 1: Baseline communication graph: a ring on $N = 6$ nodes. For this graph $\lambda_2(L) = 1$ and $\lambda_{\max}(L) = 4$, which with the unit consensus gain $k = 1$ yields a reference critical delay $\tau_{\text{crit}} = \pi/8 \approx 0.3927$ s.

Because $L\mathbf{1} = 0$, the average $\bar{p} = \frac{1}{N} \sum_i p_i$ is invariant ($\dot{\bar{p}} = -\frac{k}{N} \mathbf{1}^\top L p = 0$), and on a connected graph all remaining modes decay, so $p_i(t) \rightarrow \bar{p}(0)$ for all i : the agents rendezvous at the centroid. The decay rates are exactly $k\lambda_2, \dots, k\lambda_N$, which is why λ_2 sets the slowest mode.

3.2 Introducing the transport delay

Suppose now that the neighbour information used at time t was generated at time $t - \tau$. The *full-state* delayed protocol replaces p on the right-hand side of (4) by its delayed value:

$$\dot{p}(t) = -kLp(t - \tau). \quad (5)$$

This is a linear time-invariant delay-differential equation. Its stability is governed by the spectrum of the associated characteristic equation, which we now diagonalise.

3.3 Modal decomposition

Since L is real symmetric, it admits an orthonormal eigenbasis: there is an orthogonal matrix $V = [v_1, \dots, v_N]$ such that $V^\top L V = \text{diag}(\lambda_1, \dots, \lambda_N)$. Projecting (5) onto this basis through the modal coordinates $z = V^\top p$ decouples the system into N scalar delay equations, one per eigenvalue:

$$\dot{z}_i(t) = -k\lambda_i z_i(t - \tau), \quad i = 1, \dots, N. \quad (6)$$

The mode associated with $\lambda_1 = 0$ gives $\dot{z}_1 = 0$: it is the (conserved) average and plays no role in stability. Every other mode is a scalar equation of the form $\dot{z} = -az(t - \tau)$ with $a = k\lambda_i > 0$. The full system is stable if and only if *each* of these scalar modes is stable; the most demanding one will turn out to be the mode with the largest λ_i , i.e. $\lambda_{\max}$.

3.4 Characteristic equation and the stability boundary

Seek solutions of the scalar mode $\dot{z} = -az(t - \tau)$ of the form $z(t) = e^{st}$. Substituting gives the transcendental characteristic equation

$$s = -ae^{-s\tau}, \quad a = k\lambda_i > 0. \quad (7)$$

For $\tau = 0$ this reduces to $s = -a < 0$: the mode is stable, consistent with the delay-free analysis. As τ grows the roots $s(\tau)$ move continuously in the complex plane. The system loses stability precisely when a root first reaches the imaginary axis, i.e. when a purely imaginary root $s = j\omega$ appears. Substituting $s = j\omega$ into (7),

$$j\omega = -ae^{-j\omega\tau} = -a(\cos \omega\tau - j\sin \omega\tau). \quad (8)$$

Separating real and imaginary parts yields the two conditions

$$\text{(real)} \quad 0 = -a \cos \omega \tau, \quad (9)$$

$$\text{(imaginary)} \quad \omega = a \sin \omega \tau. \quad (10)$$

Since $a > 0$, equation (9) forces $\cos \omega \tau = 0$, hence $\omega \tau = \pi/2 (+m\pi)$. The first crossing—the one that determines the stability margin—corresponds to $\omega \tau = \pi/2$. Taking $\omega > 0$, equation (10) then gives $\omega = a \sin(\pi/2) = a$. Combining the two,

$$\omega = a, \quad \omega \tau = \frac{\pi}{2} \implies \tau = \frac{\pi}{2a} = \frac{\pi}{2k\lambda_i}. \quad (11)$$

Thus the mode with eigenvalue λ_i becomes marginally stable at delay $\pi/(2k\lambda_i)$, and is unstable for larger delays. The whole system is stable only as long as *all* modes are stable, i.e. up to the *smallest* of these per-mode thresholds, which is attained by the *largest* eigenvalue. Therefore the critical delay of the networked system is

$$\tau_{\text{crit}} = \frac{\pi}{2k\lambda_{\max}(L)} \quad (12)$$

This expression is the direct gain-scaled version of the classical uniform-delay consensus condition of Olfati-Saber and Murray [1]. In their unit-gain formulation, $\dot{x}(t) = -Lx(t - \tau)$, average consensus on a fixed connected undirected graph is achieved if and only if $0 \leq \tau < \pi/(2\lambda_{\max}(L))$. Introducing the scalar gain k replaces each Laplacian eigenvalue λ_i by $k\lambda_i$, which gives exactly (12).

Proposition 1 (Critical delay). *Consider the delayed consensus system (5) on a connected undirected graph with gain $k > 0$. All non-average modes are asymptotically stable if $0 \leq \tau < \tau_{\text{crit}}$, with τ_{crit} given by (12), and at least one mode is marginally stable at $\tau = \tau_{\text{crit}}$; for $\tau > \tau_{\text{crit}}$ the mode associated with $\lambda_{\max}$ is unstable.*

Remark 1 (Role of τ_{crit} in the nonholonomic system). *The critical delay τ_{crit} in (12) is derived for the linear delayed consensus dynamics of single-integrator agents. Therefore, it is an exact stability threshold for the underlying full-state delayed consensus layer, not for the complete nonlinear unicycle model. In the unicycle case, the desired consensus velocity must be converted into admissible inputs (v_i, ω_i) , and this introduces heading dynamics, projection effects, saturations and other nonlinearities.*

Consequently, in this report τ_{crit} is used as a theoretical baseline rather than as a strict necessary and sufficient stability condition for the nonholonomic closed-loop system. Its value remains important because the unicycle controller is built around the same consensus vector field: when τ/τ_{crit} approaches or exceeds one, the delayed communication layer tends to generate oscillatory or divergent behaviours, which are then reflected in the nonholonomic trajectories. The simulations should therefore be read as a numerical validation of how the theoretical delay margin of the consensus layer manifests itself in the more realistic unicycle rendezvous problem.

Remark 2 (Physical reading of the threshold). *The product $k\lambda_{\max}$ has the units of a rate (inverse time); it is the fastest closed-loop rate the network tries to impose. Its reciprocal $1/(k\lambda_{\max})$ is the corresponding time constant, and the delay margin (12) is a quarter period ($\pi/2$ in phase) of the oscillation that this fastest mode would sustain at the boundary. Intuitively, a delay equal to a quarter of the natural oscillation period turns the corrective action into a destabilising, out-of-phase action: the agent keeps reacting to where its neighbours were, overshooting and feeding an oscillation that grows without bound.*

3.5 Design consequences

Equation (12) exposes the two levers that set the delay margin.

Gain. The margin is inversely proportional to k : a more aggressive controller converges faster when there is no delay but tolerates *less* delay. Halving k doubles τ_{crit} . This trade-off between nominal speed and delay robustness is verified numerically in Section 6.4.

Topology. The margin is inversely proportional to $\lambda_{\max}(L)$. Denser graphs and larger edge weights raise $\lambda_{\max}$ and hence *shrink* the delay margin. A complete graph K_N with unit weights has $\lambda_{\max} = N$, so its margin decreases as the team grows, whereas a sparse path graph has the smallest $\lambda_{\max}$ among connected topologies on the same nodes and therefore the largest margin. Connectivity (λ_2) speeds up convergence but it is $\lambda_{\max}$, not λ_2 , that limits the delay.

4 The two delay models and the role of τ_{crit}

The derivation above used the *full-state* model (5), in which *both* the agent’s own state and its neighbours’ states enter delayed. A second, often more realistic, model delays *only the information that physically travels over the network*—the neighbours’ states—while each agent uses its *own* current position without delay (it is locally available). This *neighbour-only* protocol is

$$u_i(t) = -k \sum_{j \in \mathcal{N}_i} a_{ij} (p_i(t) - p_j(t - \tau)), \quad (13)$$

or, in stacked form, $\dot{p}(t) = -k(Dp(t) - Ap(t - \tau))$. The undelayed term $-kDp(t)$ adds instantaneous damping that the full-state model lacks, which substantially enlarges the delay margin. The full-state threshold (12) therefore acts as a *conservative reference*: it is exact for (5), and for (13) the true margin is larger, so τ_{crit} remains a meaningful and safe yardstick against which to normalise all delays in the experiments.

The role of τ_{crit} can thus be summarised as follows. It is the *phase margin of the linear consensus network expressed as a time*: a single number, computable from the gain and the graph spectrum alone, that predicts when delayed cooperation flips from stabilising to destabilising in the reference model. In the nonholonomic unicycle system it is not used as an exact boundary, but as the theoretical scale with which all delays are compared. All experiments below therefore report the dimensionless ratio τ/τ_{crit} , which makes results comparable across graphs of very different size and density and clarifies how the linear delay margin appears in the nonlinear unicycle simulations.

5 From the consensus field to the unicycle

The protocols (5) and (13) produce, at each instant, a desired Cartesian velocity $u_i \in \mathbb{R}^2$ for agent i . A single integrator could apply it directly; a unicycle cannot, because of the nonholonomic constraint. Two controllers are used to reconcile u_i with (1).

Vector-field tracking controller. The desired velocity is treated as a reference to be tracked in direction and magnitude. With heading error $e_\theta = \text{atan2}(u_{i,y}, u_{i,x}) - \theta_i$ wrapped to $(-\pi, \pi]$, the inputs are

$$v_i = k_v \|u_i\| \max\{0, \cos e_\theta\}, \quad \omega_i = k_\theta e_\theta. \quad (14)$$

The robot drives forward only when it roughly faces the target direction (the $\max\{0, \cos e_\theta\}$ gate) and steers proportionally to the heading error.

Force-projection (“paper”) controller. Inspired by potential-field rendezvous laws for nonholonomic agents, the vector u_i is interpreted as an interaction force and is *projected* onto the admissible direction of motion t_i :

$$v_i = k_v u_i^\top t_i, \quad \omega_i = k_\theta e_\theta. \quad (15)$$

Only the component of the force along the current heading produces translation; the orthogonal component is rejected and handled by steering. This is the default controller and the one most faithful to the reference papers.

In both cases the heading dynamics and the saturation are genuine nonlinearities, so the linear threshold (12) is no longer exact for the unicycle. This is the main reason why τ_{crit} is interpreted as a theoretical

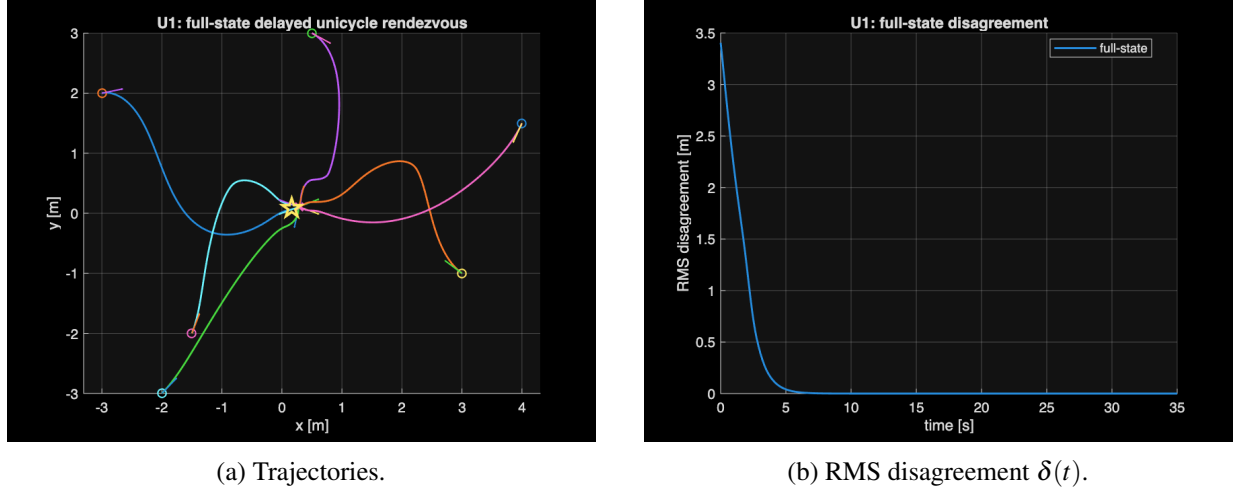


Figure 2: Baseline full-state rendezvous at $\tau = 0.3 \tau_{\text{crit}}$. The six robots (arrows show initial and final headings) converge to a common point; the disagreement decays monotonically to numerical zero.

reference for the nonholonomic system, rather than as a formal stability theorem for the complete closed loop. It remains, however, an excellent organising quantity: as the experiments show, behaviour changes *character* in the neighbourhood of $\tau/\tau_{\text{crit}} \approx 1$, even though the precise crossing is shifted by the nonlinearity.

6 Numerical validation

All experiments use the parameters $k = 1$, $k_v = 1$, $k_\theta = 3$, $v_{\text{max}} = 1.5$, $\omega_{\text{max}} = 4$, integration step $dt = 0.02$ s, on the ring graph of Figure 1 unless stated otherwise. Convergence is measured by the root-mean-square (RMS) disagreement

$$\delta(t) = \sqrt{\frac{1}{N} \sum_i \|p_i(t) - \bar{p}(t)\|^2}, \quad (16)$$

the spatial spread of the team about its centroid; rendezvous corresponds to $\delta(t) \rightarrow 0$.

6.1 Baseline rendezvous

With a sub-critical delay $\tau = 0.3 \tau_{\text{crit}}$, the full-state unicycle team rendezvouses cleanly (Figure 2). The recorded final disagreement is $\delta_\infty \approx 8.3 \times 10^{-9}$, with a convergence time of 6.22 s and a final centroid at (0.284, 0.108)—close to the centroid of the initial positions, as the theory predicts for small delay.

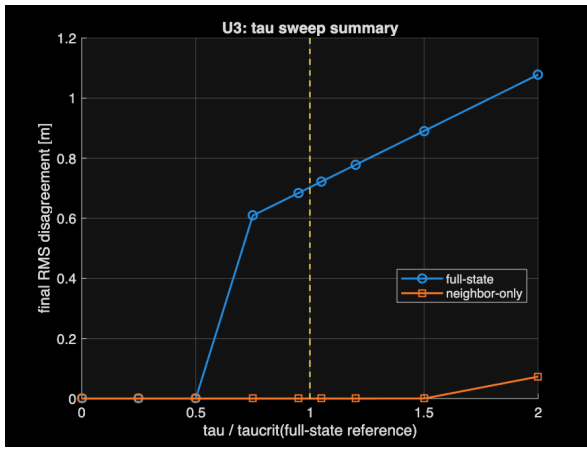
6.2 Delay sweep: the threshold in action

The single most important experiment sweeps τ from 0 to $2 \tau_{\text{crit}}$ for both delay models. Table 1 reports the final disagreement and convergence time, and Figure 3 visualises the contrast.

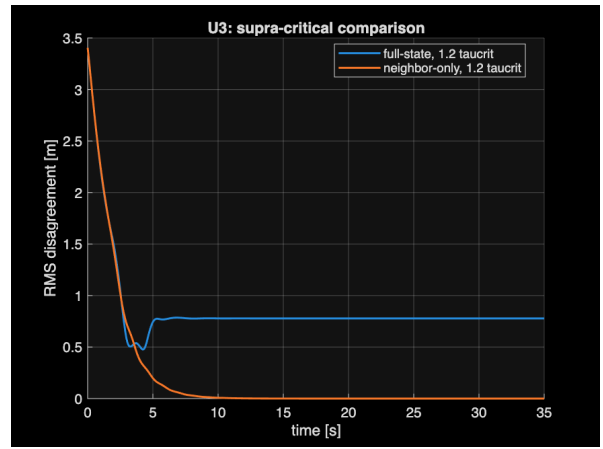
Three observations stand out. First, the **full-state model collapses across the predicted region**: it converges for $\tau \leq 0.5 \tau_{\text{crit}}$ and fails to rendezvous from $0.75 \tau_{\text{crit}}$ onward, with the residual disagreement growing monotonically with τ . That the practical crossing occurs somewhat below the ideal $\tau = \tau_{\text{crit}}$ is exactly the expected effect of the unicycle nonlinearity and the finite integration step: the linear τ_{crit} is an upper bound on the achievable margin, approached but not reached by the nonholonomic system. Second, the **neighbour-only model is dramatically more robust**: it still rendezvouses essentially perfectly at $\tau = 1.2 \tau_{\text{crit}}$ and only begins to degrade near $2 \tau_{\text{crit}}$, confirming the extra damping argument of Section 4. Third, within the convergent regime, **larger delay slows convergence** (the neighbour-only t_c grows from 6.8 to 10.5 s): even when stability is preserved, latency has a cost.

Table 1: Delay sweep on the ring graph. Final RMS disagreement and convergence time as a function of τ/τ_{crit} , for the two delay models. “—” denotes no convergence within the horizon.

τ/τ_{crit}	full-state		neighbour-only	
	δ_{∞}	t_c [s]	δ_{∞}	t_c [s]
0.00	9.5×10^{-9}	6.76	9.5×10^{-9}	6.76
0.25	8.7×10^{-9}	6.32	9.5×10^{-9}	7.30
0.50	6.5×10^{-9}	5.88	9.6×10^{-9}	7.90
0.75	0.609	—	9.6×10^{-9}	8.52
0.95	0.684	—	9.7×10^{-9}	9.02
1.05	0.722	—	9.8×10^{-9}	9.28
1.20	0.778	—	9.8×10^{-9}	9.64
1.50	0.890	—	1.4×10^{-4}	10.48
2.00	1.078	—	0.073	—



(a) Final disagreement vs τ/τ_{crit} (log scale).



(b) Disagreement curves at $\tau = 1.2 \tau_{\text{crit}}$.

Figure 3: Delay sweep, full-state vs neighbour-only. (a) The full-state curve jumps by orders of magnitude as τ crosses the threshold (dashed line at $\tau/\tau_{\text{crit}} = 1$), while the neighbour-only curve stays near zero far beyond it. (b) At $1.2 \tau_{\text{crit}}$ the full-state mode saturates at a large residual while the neighbour-only team still converges.

6.3 Topology

Equation (12) predicts that the delay margin is set by λ_{max} , hence by the density of the graph. Figure 4 compares path, ring, star and complete topologies at a common relative delay $\tau = 0.5 \tau_{\text{crit}}$ (each graph normalised by its *own* τ_{crit}). The denser graphs (star, complete) have the largest λ_{max} and therefore the smallest absolute margin; the sparse path has the smallest λ_{max} and the largest margin. At equal τ/τ_{crit} all connected topologies rendezvous, but their convergence times order themselves according to algebraic connectivity λ_2 , the denser graphs being faster.

6.4 Gain versus delay margin

To isolate the role of k we vary the consensus gain and, for each value, run at a delay fixed at $0.8 \tau_{\text{crit}}(k)$, i.e. at a constant *relative* delay. Since $\tau_{\text{crit}} \propto 1/k$, increasing k shrinks the absolute admissible delay in exact inverse proportion (Figure 5). This is the practical statement of the speed/robustness trade-off: one cannot make the network arbitrarily fast and arbitrarily delay-tolerant at the same time.

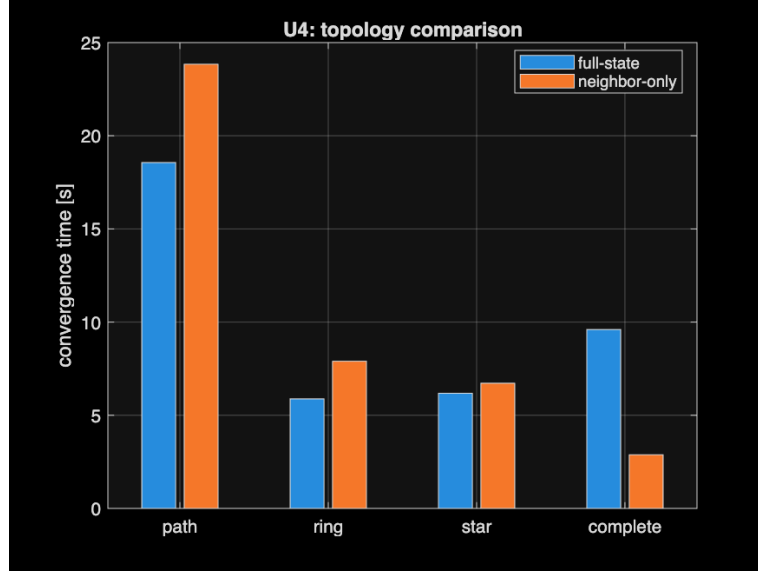


Figure 4: Topology comparison at $\tau = 0.5 \tau_{\text{crit}}$ (each graph uses its own τ_{crit}). Convergence time per topology for the two delay models; denser graphs converge faster but, in absolute terms, tolerate a smaller delay because of their larger λ_{max} .

6.5 Disconnected graph

When the graph is not connected, $\lambda_2 = 0$ and global rendezvous is impossible: the average is conserved *within each component* but not across components. Figure 6 shows a graph made of three disjoint pairs; each pair rendezvous to its own local meeting point, producing three clusters rather than one. This is the spatial signature of $\lambda_2 = 0$ and a useful negative control: it confirms that connectivity, not merely the control law, is what enables global agreement.

7 Extensions

The same delayed-consensus core is reused, with additional terms, to cover a range of more realistic coordination tasks. We highlight four representative ones.

7.1 Leader–follower with a moving target

One robot is designated leader and is driven to track a moving reference $r(t) = 1.8(\cos 0.18t, \sin 0.18t)$, while the others follow through the delayed neighbour-only consensus. The team forms a cohesive group that trails the leader along its circular path (Figure 7); the follower spread settles to a small steady value, showing that delayed consensus is compatible with a non-stationary objective.

7.2 Rigid formation

Applying consensus to *shifted* coordinates $q_i = p_i - \rho_i$, where ρ_i are prescribed offsets, turns rendezvous into a formation: the team agrees on the q_i , hence the p_i settle on the desired geometric pattern about an anchor point (Figure 8). The delay analysis carries over unchanged, since the offset is a constant exogenous term.

7.3 Obstacle and inter-robot collision avoidance

Continuous repulsive terms are added for a circular obstacle and for pairwise robot proximity. The team rendezvous while bending its paths around the obstacle and maintaining a positive minimum inter-robot

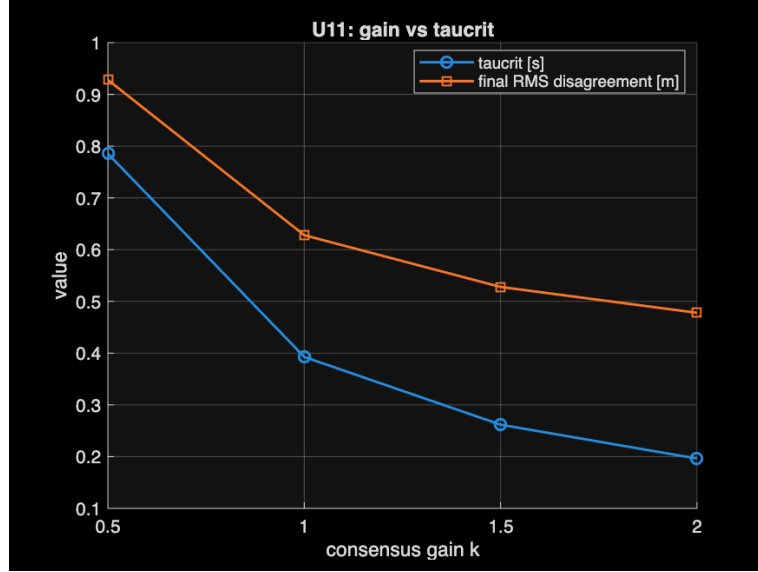


Figure 5: Effect of the consensus gain k . The critical delay τ_{crit} (left axis) falls as $1/k$, confirming (12); the final disagreement stays small because each run is kept at the same relative delay $0.8 \tau_{\text{crit}}(k)$.

clearance (Figure 9). The avoidance terms are local and vanish away from the danger zones, so they do not affect the delayed-consensus behaviour in free space.

7.4 Switching graph and time-varying delay

Two further stress tests probe the theory’s assumptions. In the first, the graph is rebuilt at every step from the robots’ positions (a proximity graph), so the topology—and hence $\lambda_2(L(t))$ and $\lambda_{\max}(L(t))$ —changes during the motion (Figure 10). The team still converges as long as the graph remains connected often enough, and the recorded $\lambda_2(t)$ stays positive. In the second, the delay itself is made time-varying, $\tau(t) = \tau_{\text{crit}} \beta (1 + \alpha \sin 2\pi f t)$, including a profile whose peak crosses τ_{crit} (Figure 10). The neighbour-only model tolerates excursions above the nominal threshold, consistent with its larger true margin, whereas the full-state model is sensitive to them.

8 Conclusions

We derived, from a frequency-domain analysis of the linear full-state delayed consensus protocol, the critical communication delay $\tau_{\text{crit}} = \pi / (2k\lambda_{\max}(L))$, and we explained its role as the time expression of the network’s phase margin. This quantity is exact for the single-integrator consensus layer, where it separates convergent from divergent cooperation, and it makes transparent the two design levers—gain and topology density—showing that aggressiveness and delay robustness are inversely related.

Transferring the consensus field to nonholonomic unicycles through a force-projection controller, we used τ_{crit} as a theoretical baseline rather than as an exact stability condition for the complete nonlinear system. The delay sweep confirms that this baseline is still informative: the full-state unicycle protocol loses rendezvous as τ approaches the linear threshold (slightly earlier, as expected from the unicycle nonlinearity), while the neighbour-only protocol—thanks to its instantaneous self-damping—remains robust well beyond it. Topology, gain, and disconnected-graph experiments confirm the dependence on $\lambda_{\max}$ and λ_2 , and a suite of extensions (leader–follower, rigid formations, obstacle avoidance, switching graphs and time-varying delays) shows that the delayed-consensus core extends gracefully to realistic coordination tasks. The overarching practical message is to normalise every delay by τ_{crit} , to interpret it as the delay margin of the underlying consensus layer, and, whenever the architecture permits, to prefer the neighbour-only protocol for its substantially larger delay margin.

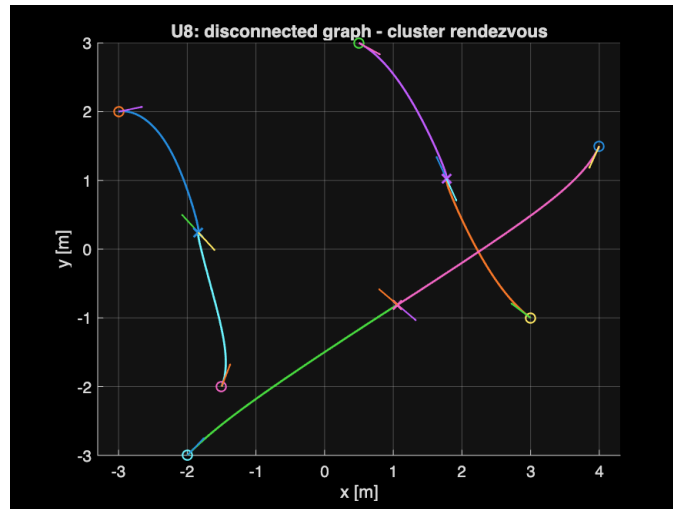


Figure 6: Disconnected graph (three disjoint pairs). Each connected component reaches its own local rendezvous point; no global agreement is possible because $\lambda_2 = 0$.

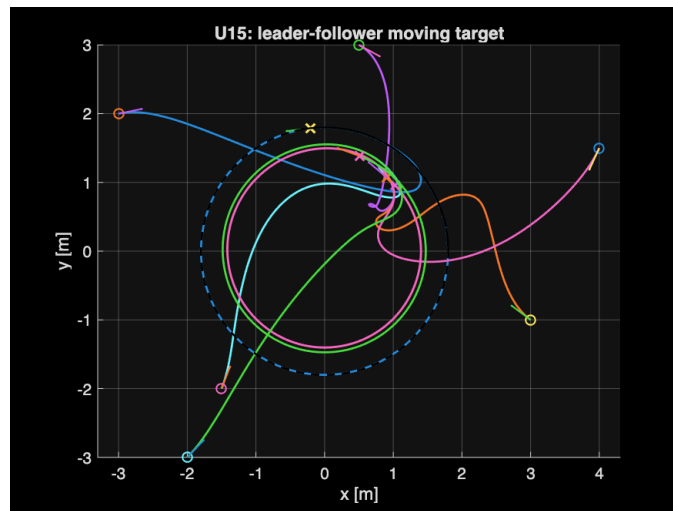


Figure 7: Leader–follower with a moving target. The leader (driven by an external reference) sweeps a circle; the followers track it cohesively through delayed consensus.

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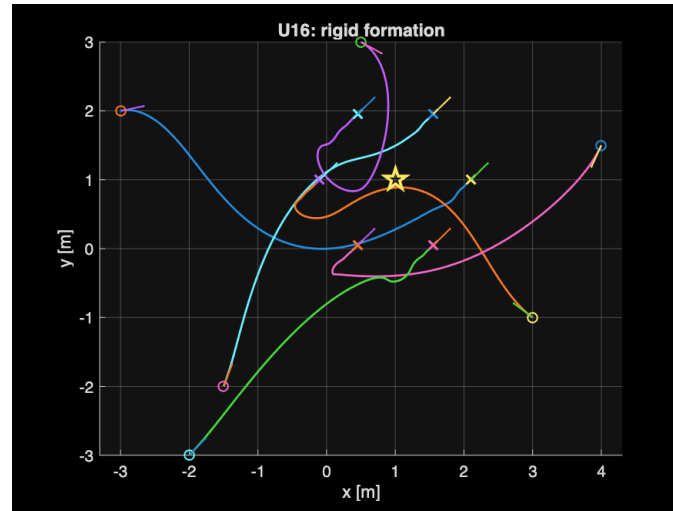


Figure 8: Rigid formation. Consensus on offset coordinates drives the robots onto a polygonal formation centred at the anchor point (star marker).

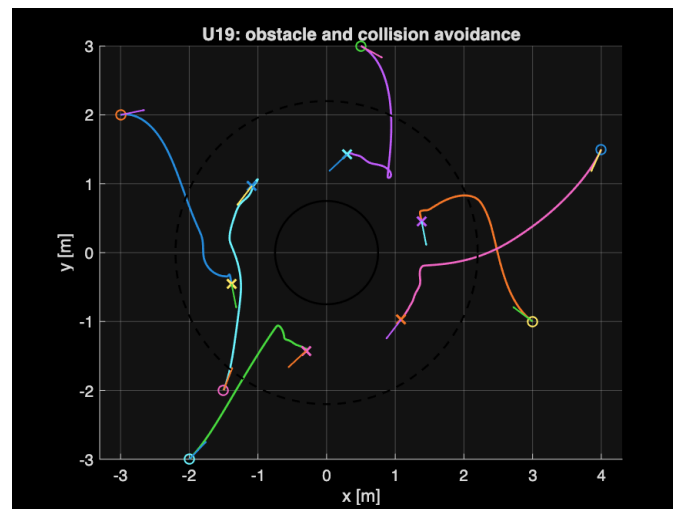
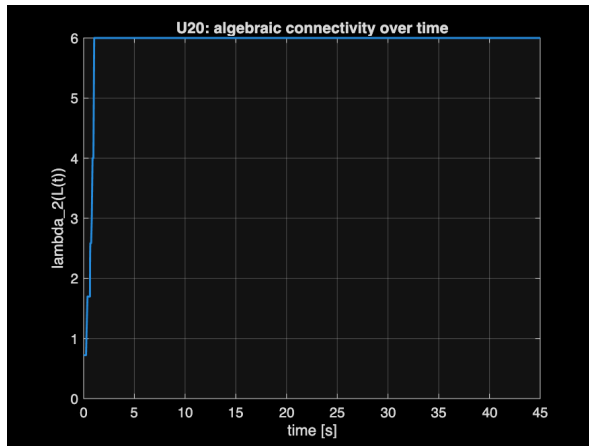
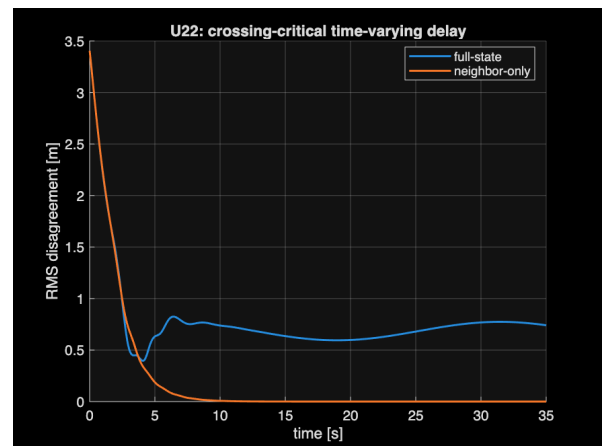


Figure 9: Obstacle and collision avoidance. Solid and dashed circles mark the obstacle and its influence region; trajectories deflect smoothly around it while the team still rendezvous.



(a) Algebraic connectivity $\lambda_2(L(t))$ of the switching proximity graph.



(b) Disagreement under a delay that crosses τ_{crit} .

Figure 10: Robustness to a time-varying network. (a) As the proximity graph switches, its algebraic connectivity varies but stays positive, preserving convergence. (b) Under a sinusoidal delay peaking above τ_{crit} , the neighbour-only protocol remains well behaved.